Marwadi University, Marwadi Chandarana Group, NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:Simulate cross correlation and autocorrelation on discrete time signals.**	
Experiment No:3	**Date:**	**Enrollment No:**92400133089

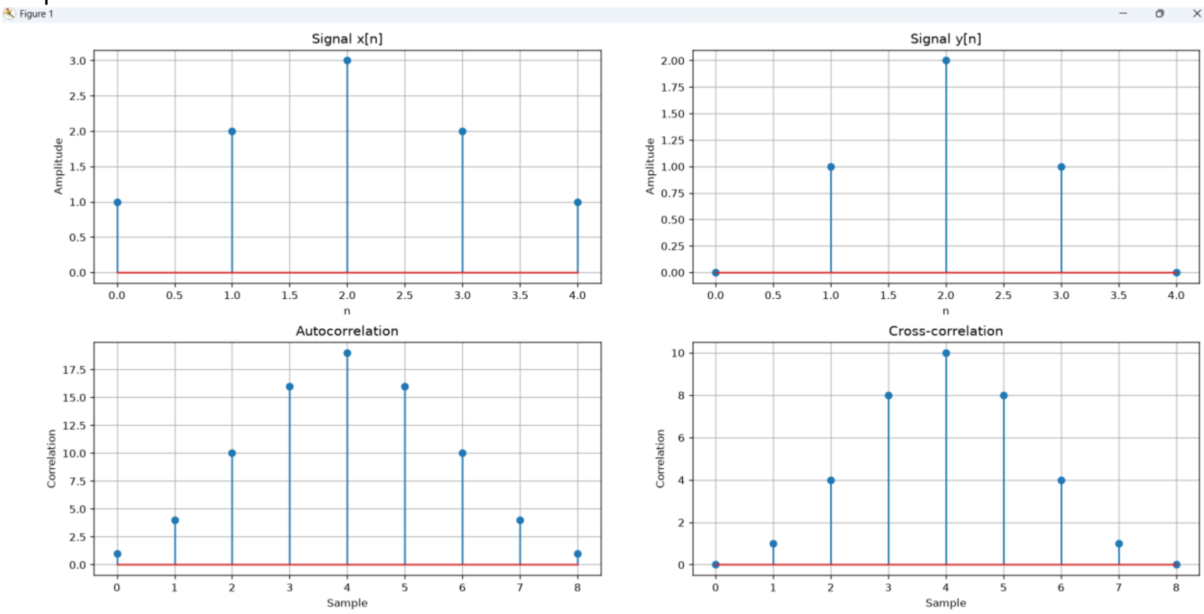
Aim:Simulate cross correlation and autocorrelation on discrete time signals.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Discrete-time signals
x = np.array([1, 2, 3, 2, 1])
y = np.array([0, 1, 2, 1, 0])

# Autocorrelation
Rxx = np.correlate(x, x, mode='full')

# Cross-correlation
Rxy = np.correlate(x, y, mode='full')

# Plot
plt.figure(figsize=(10, 8))

plt.subplot(2, 2, 1)
plt.stem(x)
plt.title("Signal x[n]")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.grid()

plt.subplot(2, 2, 2)
plt.stem(y)
plt.title("Signal y[n]")
plt.xlabel("n")
plt.ylabel("Amplitude")
plt.grid()

plt.subplot(2, 2, 3)
plt.stem(Rxx)
plt.title("Autocorrelation")
plt.xlabel("Sample")
plt.ylabel("Correlation")
plt.grid()
```

```
plt.subplot(2, 2, 4)
plt.stem(Rxy)
plt.title("Cross-correlation")
plt.xlabel("Sample")
plt.ylabel("Correlation")
plt.grid()

plt.tight_layout()
plt.show()
```

output:



observation:

1. Autocorrelation compares a signal with itself.
2. Cross-correlation compares two different signals.
3. Autocorrelation usually has its maximum at zero shift.
4. Cross-correlation shows how similar the two signals are at different shifts.
5. Autocorrelation is useful for finding repetition/patterns, while cross-correlation is useful for finding similarity between signals.

Conclusion:

Implementation done of cross and auto correlation , by experiment I want to conclude that auto correlation compares a signal with itself while Cross-correlation compares two different signals.

Github link: https://github.com/Darshit1009/DSIP